WORKED SOLUTION

STEP I 1990, Question 5

Pure • Partial Fractions and Curve Sketching

STEP I 1990, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q5

The key idea

In (i) the quadratic factorises, so partial fractions turn the integral into a difference of logarithms. In (ii) the huge powers look frightening, but $1760 = 8 \times 220$, so the function is really $u^8 - u + q$ with $u = x^{220} \geq 0$. The shape of the graph comes from the stationary points, and the number of roots depends only on where the horizontal axis sits relative to the local maximum and the two minima.

Solution

Part (i)

Since $6 \times 15 = 90 = 9 \times 10$ and $9 + 10 = 19$, the quadratic factorises:

$$6x^2 + 19x + 15 = (2x+3)(3x+5).$$

Write

$$\frac{1}{(2x+3)(3x+5)} = \frac{A}{2x+3} + \frac{B}{3x+5}, \qquad \text{so} \qquad 1 = A(3x+5) + B(2x+3).$$

Putting $x = -\frac{3}{2}$ gives $\frac{1}{2}A = 1$, so $A = 2$; putting $x = -\frac{5}{3}$ gives $-\frac{1}{3}B = 1$, so $B = -3$. Both $2x+3$ and $3x+5$ are positive on $[1,3]$, so no modulus signs are needed:

$$\begin{aligned}\int_1^3 \frac{\mathrm{d}x}{6x^2+19x+15} &= \int_1^3 \left(\frac{2}{2x+3} - \frac{3}{3x+5}\right)\mathrm{d}x = \Big[\ln(2x+3) - \ln(3x+5)\Big]_1^3 \\ &= \ln\frac{9}{14} - \ln\frac{5}{8} = \ln\frac{72}{70} = \ln\frac{36}{35}.\end{aligned}$$

Part (ii)

Symmetry and end behaviour. Both powers are even, so f is an even function and its graph is symmetric in the y-axis. For large $|x|$ the term x^{1760} dominates, so $\mathrm{f}(x) \to +\infty$ as $x \to \pm\infty$. Also $\mathrm{f}(0) = q$ and $\mathrm{f}(\pm 1) = q$.

Stationary points. Differentiating,

$$\mathrm{f}'(x) = 1760x^{1759} - 220x^{219} = 220\,x^{219}(8x^{1540} - 1).$$

This vanishes at $x = 0$ and where $x^{1540} = \frac{1}{8}$, that is at $x = \pm a$ with

$$a = 8^{-1/1540} \approx 0.9987.$$

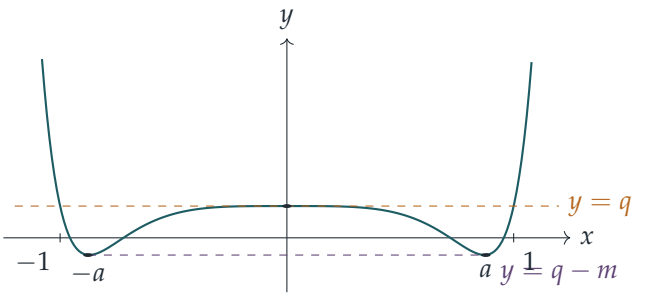
For $x > 0$, $\mathrm{f}'(x)$ has the sign of $8x^{1540} - 1$: negative on $(0, a)$ and positive for $x > a$. Since x^{219} is an odd power, the signs are reversed for $x < 0$. So f decreases on $(-\infty, -a)$, increases on $(-a, 0)$, decreases on $(0, a)$ and increases on (a, ∞): there is a local maximum q at $x = 0$ and two equal minima at $x = \pm a$.

Ask yourself How do we find the minimum value without wrestling with $8^{-1/1540}$?

Work with $u = x^{220}$. At $x = \pm a$ we have $x^{1540} = u^7 = \frac{1}{8}$, so $u = 8^{-1/7}$ and $u^8 = u \cdot u^7 = \frac{1}{8}u$. Hence

$$\mathrm{f}(\pm a) = u^8 - u + q = q - \tfrac{7}{8}u = q - m, \qquad \text{where} \qquad m = \tfrac{7}{8} \cdot 8^{-1/7} = 7 \cdot 2^{-24/7} \approx 0.650.$$

The sketch. The graph is a "W": it starts high, falls to the minimum $q - m$ at $x = -a$, rises to the local maximum q at the origin, falls to the minimum $q - m$ at $x = a$ and then rises steeply. Because the powers are so large, the real graph is extremely flat: x^{220} is tiny unless $|x|$ is very close to 1, so $\mathrm{f}(x) \approx q$ on most of $(-1, 1)$, with two narrow dips just inside $x = \pm 1$ (where the graph returns to height q) and near-vertical walls outside. The picture below exaggerates the dips so the features can be seen.



Counting roots. The roots of $\mathrm{f}(x) = 0$ are where the graph meets the x-axis, so we compare 0 with the local maximum q and the minimum value $q - m$.

- $q > m$: the minimum value $q - m$ is positive, so $\mathrm{f} > 0$ everywhere: **no roots**.
- $q = m$: the graph touches the axis only at the minima: **two roots**, $x = \pm a$.
- $0 < q < m$: now $q - m < 0 < q$. On each of the four intervals of monotonicity, $(-\infty, -a)$, $(-a, 0)$, $(0, a)$, (a, ∞), f is continuous and strictly monotonic and passes from a positive value to a negative one or vice versa, so it has exactly one root on each: **four roots**.

- $q = 0$: $\mathrm{f}(x) = x^{220}(x^{1540} - 1)$, which vanishes exactly at $x = 0$ and $x = \pm 1$: **three roots**.
- $q < 0$: the local maximum $\mathrm{f}(0) = q$ is negative, so $\mathrm{f} < 0$ on the whole of $[-a, a]$, and f is strictly monotonic from a negative value to $+\infty$ on each of (a, ∞) and $(-\infty, -a)$: **two roots**.

So the possible numbers of distinct roots are 0, 2, 3 and 4, with

$$\begin{array}{ll} 0 \text{ roots}: & q > m, \\ 2 \text{ roots}: & q < 0 \text{ or } q = m, \\ 3 \text{ roots}: & q = 0, \\ 4 \text{ roots}: & 0 < q < m, \end{array} \qquad m = \tfrac{7}{8} \cdot 8^{-1/7} = 7 \cdot 2^{-24/7} \approx 0.650.$$

Checks

Simpson's rule on the integrand in (i) gives 0.0281709, matching $\ln \frac{36}{35} = 0.0281709$. For (ii), a fine numerical search for the minimum of $x^{1760} - x^{220}$ near $x = 1$ gives -0.650123 at $x \approx 0.99865$, in agreement with $-m$ and a. Counting sign changes of f on a fine grid for $q = 1,\ 0.5,\ 0.1,\ 0.001,\ 0,\ -0.01,\ -5$ gives $0, 4, 4, 4, 3, 2, 2$ roots, as the table predicts. The case $q = 0$ is a good sanity check on the sketch: the graph must pass through the origin and through $(\pm 1, 0)$.

Common slips

- Forgetting the boundary cases $q = 0$ (three roots, because the local maximum sits on the axis) and $q = m$ (two repeated roots at the minima).
- Claiming two roots for $q < 0$ "by symmetry" without checking that nothing happens between $-a$ and a: it is the fact that the local maximum q is negative which rules out extra roots there.
- Sketching a gently curved "W". With powers this large the graph is almost flat at height q for $|x| < 1$; it is worth saying so even if the sketch exaggerates the dips.

Beyond STEP

Counting roots by locating the stationary values is a first taste of the intermediate value theorem and monotonicity arguments made rigorous in a first analysis course; the substitution $u = x^{220}$ also shows how x^n behaves as $n \to \infty$, converging to 0 on $(-1, 1)$ but not uniformly. Abbott's *Understanding Analysis* treats both continuity and uniform convergence with care and readability.

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